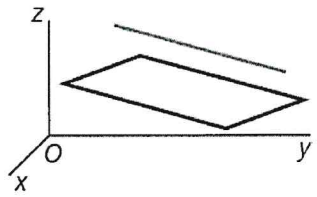
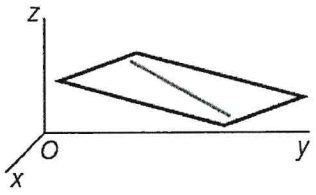
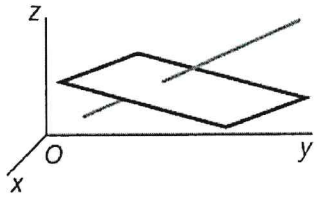
$\vec{n}$ = normal defines plane

$\vec{m}_\ell$ = direction vector of line defines line

Intersection of a Line and a Plane

		
Line is parallel to the plane	Line lies in the plane	Line intersects the plane
$\vec{n} \cdot \vec{m}_\ell = 0$	$\vec{n} \cdot \vec{m}_\ell = 0$	$\vec{n} \cdot \vec{m}_\ell \neq 0$.
No Solution	Infinite # of Solution	One Solution
$\emptyset t = \#$ not possible	$\emptyset t = \emptyset$ t can be anything	$t = \#$ t only can be the #

To determine the solution(s), write the equation of a line in PARAMETRIC form and sub into the CARTESIAN equation of the plane.

Example 1: Determine the intersection of the given line and plane.

$\vec{r} = (2,3,-1) + t(2,1,-5), t \in \Re$ and $4x - 3y + 2z - 12 = 0$

Check

$\vec{n} \cdot \vec{m}_\ell = (4,-3,2) \cdot (2,1,-5)$ $\therefore$ intersect at one point!

$= 8 - 3 - 10$

$\neq 0$.

Line

$x = 2 + 2t$
$y = 3 + t$
$z = -1 - 5t$

} sub into Plane.

$4(2+2t) - 3(3+t) + 2(-1-5t) - 12 = 0$

$8 + 8t - 9 - 3t - 2 - 10t - 12 = 0$

$-5t = 15$

$t = -3.$

one solution (as expected)

P.O.I.

$\therefore$ $x = 2 + 2(-3) = -4$

$y = 3 + (-3) = 0$

$z = -1 - 5(-3) = 14.$

$\therefore$ P.O.I. $(-4, 0, 14)$

Example 2: Determine the intersection of the given line and plane.

$$\left.\begin{aligned} x &= 1+3t \\ y &= 2+2t \\ z &= 3 \end{aligned}\right\} \; t \in \Re \qquad \text{and} \qquad 2x-3y+4z+1=0$$

Check: $\vec{n} \cdot \vec{m_\ell} = (2,-3,4) \cdot (3,2,0)$

$= 6-6+0 = \phi$.

$\therefore$ no solution or infinite solutions

Sub line into plane

$2(1+3t) - 3(2+2t) + 4(3) + 1 = 0$

$2+6t-6-6t+12+1 = 0$

$0t = -9$

not possible

$\therefore$ No solutions. and the line & plane must be parallel & distinct.

Example 3: Determine the intersection of the given line and plane.

$$x-3 = \frac{z+2}{2},\; y = 5 \qquad \text{and} \qquad 4x+3y-2z-31=0$$

Check: $\vec{n} \cdot \vec{m_\ell} = (4,3,-2) \cdot (1,0,2)$

$= 4+0-4 = \phi$

$\therefore$ no solution or infinite solutions

Line

$x = 3+t$

$y = 5$

$z = -2+2t$

Sub into plane.

$4(3+t) + 3(5) - 2(-2+2t) - 31 = 0$

$12+4t+15+4-4t-31 = 0$

$0t = \phi$

$t \in \mathbb{R}$. (infinite solutions)

$\therefore$ Infinite solution.

Line lies in the plane and all points on the line are points of intersection

$\therefore (x,y,z) = (3+t, 5, -2+2t)$, $t \in \mathbb{R}$.